5 Ionic Equilibrium

SOLUTION (QR CODE)

6. (B) Given: conc. of $H_3O^+ = 10^{-6}$ mol/L

Ionization reaction of H_2O is:

$H_2O \rightleftharpoons H^+ + OH^-$

$K_W = [H^+][OH^-]$

$[H^+] = [OH^-] = 10^{-6}$ mol/L

($\because$ it mentioned that it is pure water)

$K_W = [H^+][OH^-] = 10^{-6} \times 10^{-6} \Rightarrow 10^{-12}$

15. $[H^+]_{acid} = 10^{-8}$ M

$[H^+]_{acid} < 10^{-6}$ M

Pani ki bat sunni padegi

$$[H^+]_{Total} = [H^+]_{acid} + [H^+]_{H_2O}$$

$= 10^{-8} + x$

$$K_w = [H^+][OH^-]$$

$$10^{-14} = (10^{-8} + x)x$$

$x = ?$

$$pH = -\log(10^{-8} + x)$$

pH = 6.98

19. **(A) pH = –log [H⁺]**

$[H^+]$ = Antilog (–7.3) = 5.0×10^{-8}M

$$[OH^-] = \frac{10^{-14}}{(5.0 \times 10^{-8})} = 2 \times 10^{-7}\ M$$

$$[OH]^- = \frac{n}{V}$$

$n = 2 \times 10^{-7} \times 20 \times 10^3 = 40 \times 10^{-4}$ mol

$$\text{Moles} = \frac{\text{Mass}}{\text{Molar mass}}$$

Mass = Moles × molar mass

$= 40 \times 10^{-4}$ mol × 40 g/mol = 0.16 g

22. First calculate M_1V_1 of H_2SO_4 as;

$$M_1V_1 \text{ of } H_2SO_4 = 2 \times 400 \times \frac{1}{200} = \frac{4}{1000}$$

$$M_2V_2 \text{ of } H_2SO_4 = 400 \times \left(\frac{1}{100}\right) = \frac{4}{1000}$$

H^+ ions from water will be neglected.

$N_1V_1 + N_2V_2 = 8 \times 10^{-3}$

$$[H^+] = \frac{8 \times 10^{-3}}{1} = 8 \times 10^{-3}$$

$pH = -\log[H^+] = -\log[8 \times 10^{-3}] = 2.1$

23. (D) pH = 3, $[H^+] = 10^{-3}$M

pH = 5, $[H^+] = 10^{-5}$M

Calculate total $[H^+]$ as:

$$\text{Total } [H^+] = \frac{10^{-5} + 10^{-3}}{2}$$

$$= \frac{.01 \times 10^{-3} + 1 \times 10^{-3}}{2}$$

$$= \frac{(1 + .01) \times 10^{-3}}{2}$$

$$[H^+] = \frac{1.01 \times 10^{-3}}{2} = 0.505 \times 10^{-3}$$

$pH = -\log[0.505 \times 10^{-3}]$

pH = 3.3

26.

HCl	+ NaOH	⟶ NaCl	+ H_2O
10	15	0	
0	5	10	

$$[OH^-]_{left} = \frac{5}{250} = \frac{1}{50} M$$

$$pOH = -\log[OH^-] = -\log\frac{1}{50} = \log 50 = 1.7$$

pH = 14 – 1.7 = 12.3

27. (D) mmol of HCl = Molarity × Volume (mL)

$$= \frac{1}{5} \times 75 = 15 \text{mmol}$$

mmol of NaOH $= \frac{1}{5} \times 25 = 5$ mmol

After neutralisation, mmol of HCl remaining = 15 – 5 = 10 mmol

$\therefore$ Molarity of HCl $= \frac{10}{100} = 0.1 = [H^+]$

pH = –log $[H^+]$ = –log (0.1) = 1

daro mat bacho, asaan hi hai

34. (C) The formula to calculate K_a is:

$K_a = C\alpha^2$

Acid is 3.2% ionized, so,

$K_a = 0.2 \times \left(\frac{3.2}{100}\right)^2 = 2.048 \times 10^{-4} = 2.1 \times 10^{-4}$

bhayi asaaan hi hai

36. (D) $[H^+] = 10^{-3}M$

Consider the dissociation of HQ as:

$$HQ \rightleftharpoons H^+ + Q^-$$

Initial concentration 0.1M 0 0

$[Q^-]$ at equilibrium = $10^{-3}M$

Hence, [HQ] = 0.1M – $10^{-3}M$

$\approx 0.1M$ [$\because 10^{-3}M \lll 0.1M$]

$K_a = \frac{[H^+][Q^-]}{[HQ]} = \frac{[10^{-3}][10^{-3}]}{[0.1]}$

$K_a = 1 \times 10^{-5}$

37. (B) Ammonia undergoes 5% ionization, so,

$[OH^-] = 0.05 \times 0.02$

$[OH^-] = 10^{-3}M$

pOH = –log$[OH^-]$

= –log$[10^{-3}]$ = 3

pH + pOH = 14

pH = 14 – 3 = 11

46. (D) For a strong acid and weak base, pH of the salt can be calculated as:

$pH = \frac{1}{2}\left[pK_w - pK_b - \log C\right]$

Substitute the values in the above expression:

$pH = \frac{1}{2}\left[14 - pK_b - \log C\right]$

$pH = 7 - \frac{5}{2} - \frac{1}{2}\left(\log 2 \times 10^{-2}\right)$, pH = 5.35

50. (D) Consider the dissociation of CH_3COONa as:

$CH_3COONa \rightarrow CH_3COO^- + Na^+$

0.1 M - -

- 0.1 M 0.1 M

Now,

$CH_3COO^- + H_2O \rightleftharpoons CH_3COOH + OH^-$

C - -

C(1 – h) Ch Ch

$\Rightarrow K_h = \frac{[CH_2COOH][OH^-]}{[CH_3COO]} = \frac{K_w}{K_a} = \frac{Ch^2}{(1-h)}$

$\Rightarrow \frac{10^{-14}}{10^{-5}} = Ch^2$

{$\because$ h is very small $\therefore$ 1 – h ≈ 1}

$\Rightarrow h = \sqrt{\frac{10^{-9}}{0.1}} = 10^{-4}$

$\therefore [OH^-] = Ch = 0.1 \times 10^{-4} = 10^{-5}$

$\Rightarrow [H^+] = 10^{-9}$, pH = – log $[H^+]$ = 9

kaafi mazedaar sawaal hai

51. (D) First write the reaction as:

$CH_3COO^- + H_2O \rightarrow CH_3COOH + OH^-$

CH_3COONa is the salt of a weak acid and a strong base.

Now, calculate $[OH^-]$ as:

$\therefore [OH^-] = C.h = C\sqrt{\left[\frac{K_h}{C}\right]}$

$= \sqrt{\frac{K_w}{K_a}}.C$

$= \sqrt{\frac{10^{-14} \times 1}{1.8 \times 10^{-5}}} = 2.35 \times 10^{-5}$

$\therefore$ pOH = 4.6289

pOH can be approximated as 4.6.

pH + pOH = 14

pH = 14 – 4.6 = 9.4

Hence, D is the correct option.

58. (D) $X^- + H_2O \rightleftharpoons HX + OH^-$

0.01 0 0

0.01 – 0.0001 0.0001 0.0001

$$K_b = \frac{[HX][OH^-]}{[X^-]}, K_b = 10^{-6}$$

pOH = – log $[OH^-]$ = –log 10^{-4}

pOH = 4, pH = 14 – 4 = 10

$$K_a = \frac{K_w}{K_b} = \frac{10^{-14}}{10^{-6}} = 10^{-8}$$

$K_a = 10^{-8}, pH = 10$

sara khel samajh ka hai

59. (A) $K_a = \frac{K_W}{K_h}$

$K_h = Ca^2$ ($C \rightarrow$ concentration of salt)

$$K_h = 0.1 \times \left(\frac{3}{100}\right)^2 (h \Rightarrow \text{degree of hydrolysis})$$

$K_h = 9 \times 10^{-5}$, $K_w = 10^{-14}$

$$K_a = \frac{10^{-14}}{9 \times 10^{-5}} = 1.1 \times 10^{-10}$$

60. (A,C) Neutral salt is formed by the mixture of strong acid and strong base. Among the given options, NaCl is a neutral salt.

Also, for CH_3COOH, NH_4OH, values of K_a and K_b are same, so, it is also a neutral salt.

samajh aya ya nahi?

65. (B) $pH = pK_a + \log \frac{[C.B]}{[Acid]}$

$$10^{-4} = [H^+] = \frac{K_a[CH_3COOH]}{[CH_3COO^-]}$$

$$\frac{[CH_3COO^-]}{[CH_3COOH]} = \frac{K_a}{10^{-4}} = \frac{1.3 \times 10^{-5}}{10^{-4}}$$

$= 1.3 \times 10^{-1} = 0.13$

67. $pH = pK_a + \log \frac{[CH_3COO^-]}{[CH_3COOH]}$

$$5.74 = 4.74 + \log \frac{[CH_3COO^-]}{[CH_3COOH]}$$

$$\log \frac{[CH_3COO^-]}{[CH_3COOH]} = 1, \frac{[CH_3COO^-]}{[CH_3COOH]} = 10$$

$[CH_3COO^-] = 10[CH_3COOH]$

$= 10 \times 1$ M = 10 M

tedha lagra hoga lekin hai nhi, bharosa karo

71. This is an acidic buffer and its formula to calculate pH is:

$$pH = pK_a + \log \frac{[salt]}{[Acid]}$$

$$4 = -\log 10^{-5} + \log \frac{[salt]}{[0.2M]}$$

$$-1 = \log \frac{[Salt]}{[0.2M]}, \text{Log}[Salt] = -1.698$$

[Salt] = 0.02 M

direct formula lagaya hai bas

77. (C) This is a basic buffer.

$pOH = pK_b + \log [salt]/[base]$

$14 - 8.26 = 4.74 + \log[NH_4Cl]/[0.2]$

$5.74 = 4.74 + \log [NH_4Cl]/[0.2]$

$1 = \log[NH_4Cl] - \log (0.2)$

$0.301 = \log [NH_4Cl]$

$[NH_4Cl] = 2$

Now, calculate amount as:

Amount of $NH_4Cl = 2 \times 53.5 = 107$ g

sawaal ki language dekh kar darna mat

86. Consider the dissociation of A_2X:

$A_2X \rightarrow 2A^+ + X^{2-}$

$K_{sp} = 4S_1^3 = 4 \times 10^{-12}$

$S_1 = 10^4$

Now, consider the dissociation of MX:

$MX \rightarrow M^+ + X^-$

$S_2 \quad S_2$

$K_{sp} = S_2^2 = 4 \times 10^{-12}$

$S_2 = 2 \times 10^{-6}$

Calculate the ratio of S_1 and S_2 as:

$$\frac{S_{A_2X}}{S_{MX}} = \frac{10^{-4}}{2 \times 10^{-6}} = 50$$

dekh toh lo sawaal ko ki kya puch raha hai?

88. (D) First write the dissociation of MX_4 as:

$MX_4 \rightleftharpoons M^{4+} + 4X^-$

$K_{sp} = [M^{4+}][4X^-]^4$

$K_{sp} = (S)(4S)^4 \Rightarrow K_{Sp} = 256\ S^5$

$$S^5 = \frac{K_{sp}}{256} \Rightarrow S = \left(\frac{K_{sp}}{256}\right)^{\frac{1}{5}}$$

90. (A) Ionisation equation is given as,

$Cr(OH)_3 \rightleftharpoons Cr^{3+} + 3OH^-$

$S \qquad 3S$

$K_{sp} = [Cr^{3+}][OH^-]^3$

$6 \times 10^{-31} = S \times (3S)^3$

$6 \times 10^{-31} = 27\ S^4$

$$S^4 = \frac{6 \times 10^{-31}}{27}$$

$$S = \left(\frac{6}{27} \times 10^{-31}\right)^{\frac{1}{4}}$$

$$= 3\left(\frac{6}{27} \times 10^{-31}\right)^{\frac{1}{4}} \quad [OH^-] = 3S$$

$$= 3\left(\frac{6}{27} \times 10^{-31}\right)^{\frac{1}{4}} M = \left(18 \times 10^{-31}\right)^{\frac{1}{4}} M$$

92. (A) The given compound is $Cr(OH)_3$, which dissociates in water as:

$Cr(OH)_3 \rightleftharpoons Cr^{3+} + 3OH^-$

The solubility product constant (K_{sp}) expression for $Cr(OH)_3$ is:

$K_{sp} = [Cr^{3+}][OH^-]^3$

If S is the molar solubility of $Cr(OH)_3$, then:

$[Cr^{3+}] = S, [OH^-] = 3S$

Substituting these in the above expression:

$K_{sp} = S(3S)^3 = S \times 27S^3 = 27S^4$

Solving for S:

$$S = \sqrt[4]{\frac{K_{sp}}{27}}$$

Given $K_{sp} = 1.6 \times 10^{-30}$, we get:

$$S = \sqrt[4]{\frac{1.6 \times 10^{-30}}{27}}$$

Arey kyu ghabra rahe ho, mushkil nahi hai

94. (C) Amount of Ag^+ required to precipitate AgCl is:

$K_{sp} = IP = [Ag^+][Cl^-] = 1.7 \times 10^{-10}$

$[Ag^+] = 1.7 \times 10^{-9}$

Now, calculate the amount of Ag^+ required to precipitate Ag_2CrO_4 is:

$K_{sp} = IP = [Ag^+]^2 [CrO_4^{-2}] = 1.9 \times 10^{-12}$

$[Ag^+] = 4.3 \times 10^{-5}$

$[Ag^+]$ required to ppt AgCl is low so AgCl will ppt 1st

isko padhkar zarur mza aagya hoga!

97. Consider the dissociation of $CdSO_4$ as:

$CdSO_4(s) \rightleftharpoons Cd^{+2}(aq) + SO_4^{2-}(aq)$

$S \qquad S$

Calculate K_{sp} as:

$K_{sp} = S^2 = 64 \times 10^{-8}$

In H_2SO_4,

$CdSO_4(s) \rightleftharpoons Cd^{2+} + SO_4^{2-}$

$S \qquad S + 10^{-2}$

$K_{sp}(CdSO_4) = S(S + 10^{-2})$

$S + 10^{-2} = 10^{-2} \because$ solubility

It can be seen that solubility is much less than 0.01 M, so,

$64 \times 10^{-8} = S \times 10^{-2}$

$S = 64 \times 10^{-6}$

100. (C) The molar solubility of Ag_2CO_3

$Ag_2CO_3 \rightleftharpoons 2Ag^+ + CO_3^{2-}$

$2S + 0.1 \quad S$ बेईमानी हमेशा करनी है

On adding 0.1 M Ag^+

$8 \times 10^{-12} = [Ag^+][CO_3^{2-}]$

$= (2S + 0.1)^2 S = 0.01S$

$S = 100 \times 8 \times 10^{-12} = 8 \times 10^{-10}$ Mol/L

102. (B) For precipitation, Q_{ip} should be greater than K_{sp}.

Consider the dissociation of $[A(OH)_2]$ as:

$[A^{2+}][OH^-]^2 > 9 \times 10^{-10}$

$[A^{+2}] = 1\ M \Rightarrow [OH^-] > 3 \times 10^{-5}\ M$

Now, consider the dissociation of $[B(OH)_3]$:

$[B^{3+}][OH^-]^3 > 27 \times 10^{-18}$

$[B^{3+}] = 1\ M \Rightarrow [OH^-] > 3 \times 10^{-6}\ M$

From above calculations, it can be said that $[B(OH)_3]$ will precipitate before $[A(OH)_2]$

103. This is a case of simultaneous solubility.

K_{sp} of CuCl > K_{sp} of AgCl

It is assumed that Cl^- will mainly come from CuCl .

$\left[Cl^-\right] = \sqrt{K_{sp}(CuCl)} = 10^{-3} M$

Now , for AgCl, $K_{sp} = 1.6 \times 10^{-7} = [Ag^+]\,[Cl^-]$

$=[Ag^+] = 1.6 \times 10^{-7}$

Hence, the value of x is '7'

104. (D) AgBr on dissociation gives Ag^+ and NO_3^- ions

This Ag^+ will combine with NH_3 and form a complex and due to this, solubility will increase drastically.

yeh badiya sawaal hai

106. (D) $AgCN_{(S)} \rightleftharpoons \underset{x}{Ag^+_{(aq)}} + \underset{x-y}{CN^-_{(aq)}}$

$\underset{x-y}{CN^-_{(aq)}} + \underset{10^{-3}}{H^+_{(aq)}} \rightleftharpoons \underset{y}{HCN_{(aq)}}$

$K_{SP} = x\,(x - y)$ ...(1)

$\frac{1}{K_a} = \frac{y}{(x-y)\,10^{-3}}$ ...(2)

Multiplying eq (1) and (2)

$\frac{K_{SP}}{K_a} = \frac{x \cdot y}{10^{-3}} = \frac{x \cdot x}{10^{-3}}$ $\left\{\frac{1}{K_a} \text{ is very large}\right.$ $\left.x - y \simeq 0\,;\, x \simeq y\right\}$

$x = \sqrt{\frac{10^{-3}\,K_{SP}}{K_a}} = 1.9 \times 10^{-5}$

107. (C) When equal volumes of two solutions are mixed, the molarity of each component decreases to half.

And, for precipitation: $Q_{SP} > K_{SP}$

For option (C)

$Q_{SP} = [A^{+2}]\,[Y^-]^2$

$= \left(\frac{2.0 \times 10^{-2}}{2}\right) \times \left(\frac{2.0 \times 10^{-2}}{2}\right)^2 = 10^{-6}$

i.e., QSP $(10^{-6}) > K_{SP}(5.2 \times 10^{-7})$

For rest other options, $Q_{SP} < K_{SP}$

108. As, $pOH = pK_b + \log\frac{[salt]}{[base]} = pK_b + \log\frac{[NH_4^+]}{[NH_3]}$

$\Rightarrow pOH = 4.745 + \log\frac{[0.10]}{[0.10]}$

$\Rightarrow pOH = 4.745$

$\Rightarrow \therefore\ pH = 14 - 4.745 = 9.255$ ...(1)

Now, on addition of 0.05 moles of HCl

NH_3	+ H^+	$\longrightarrow NH_4^+$
0.1	0.05	0.1
0.05	0	0.15

Now, applying Henderson-Hasselbalch equation:

$pOH' = 4.745 + \log\frac{[0.15]}{[0.05]}$

$pOH' = 4.745 + \log 3$

$\therefore pH' = 14 - (4.745 + \log 3)$

$pH' = 8.777$ ...(2)

$\therefore\ \Delta pH = pH - pH'$

$= 9.255 - 8.77$

$= 0.478$

i.e.; $\Delta pH = 47.8 \times 10^{-2} \sim 48 \times 10^{-2}$